

Equivalencia entre densidad y complemento con interior vacío

Proposición 1. Sea $(X, \mathcal{T})$ un espacio topológico y $D \subset X$ un subconjunto, entonces

$$D \text{ es denso en } X \iff D^c \text{ tiene interior vacío.}$$

Demostración: Veamos directamente la equivalencia por definición de interior:

$$\begin{aligned} \text{Int}(D^c) = \emptyset &\iff \neg(\exists x \in X : \exists U \in \mathcal{V}(x) : U \subset D^c) \iff \forall x \in X : \forall U \in \mathcal{V}(x) : U \not\subset D^c \\ &\iff \forall x \in X : \forall U \in \mathcal{V}(x) : U \cap D \neq \emptyset \iff \forall U \in \mathcal{T} \setminus \{\emptyset\} : U \cap D \neq \emptyset, \end{aligned}$$

pero lo la prop-con-denso/Proposición 1, esto es equivalente a que D sea denso en X . ■

Referenciado en

- cor-baire

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